

Interpretable Defense Portfolios for Synthetic Hospital Ransomware: Separating Outcome Sensitivity from Decision Stability

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Abstract—Choosing defenses from a security simulation requires knowing which results depend on its assumptions. We develop an evidence-constrained synthetic hospital ransomware model with pathway-specific controls, six decision objectives, and 160,000 confirmation executions paired across 400 profile-candidates. We separate parameter uncertainty, structural choices, outcome definitions, and efficient-set membership. Changing the outage-service threshold shifts high-capacity candidate-average event frequency by 14.34 percentage points while preserving its observed Pareto frontier. A finite interval certificate goes further: 72 of 95 observed efficient portfolios remain efficient throughout candidate-specific endpoint intervals, with five other objectives fixed. All 9 high-capacity frontier members are retained, although 7 additional candidates remain possibly efficient. Published truss and beam benchmarks provide transfer checks over 6,000 perturbations and 512 exact corner matrices; wide bounds expose when certification becomes inconclusive. Paired stratified analysis, structural sensitivity, and executable interpretation assertions complete the evaluation. The contribution is an inspectable model and a decision-stability analysis that distinguishes guaranteed retention from observed agreement. Its guarantees concern the declared objective bounds, not clinical effectiveness.

Index Terms—healthcare cybersecurity; simulation; parameter identification; Pareto efficiency; interval uncertainty; reproducibility

1. Introduction

Hospital ransomware defense involves competing goals: reducing disruption and extreme losses while controlling implementation cost and operational burden. Incident studies document disruption [1], but those observations do not identify what a particular hospital would gain from a particular combination of controls. Simulation makes such counterfactuals inspectable, provided its assumptions and decision implications are explicit.

We develop a synthetic hospital network model with pathway-specific defenses, clinical-service dependencies, and paired scenario comparisons. Its evidence framework distinguishes observed incident quantities, evidence-informed context, declared technical assumptions, and normalized decision units. This distinction prevents an adoption survey or a guidance document from being treated as an

estimate of control effectiveness. The finite experiment enumerates 400 profile-candidates and compares six objectives without collapsing them into a single preferred policy.

The central question is: *when modeled outcomes change, which portfolio decisions can still be defended?* Changing the service-outage definition can substantially change event frequency while preserving the estimated efficient set. Conversely, changing a restoration mechanism can alter that set even when parameter-based ranking checks appear reassuring. These are different forms of sensitivity, requiring different evidence.

We make three contributions. First, we provide an inspectable defense-portfolio experiment that links evidence classification to mechanisms, matched scenarios, and separate parameter and structural sensitivity analyses. Second, we derive and implement a finite interval certificate that bounds every admissible frontier between guaranteed and possible efficient sets, with competing portfolios as failure witnesses. Applied to the endpoint interval, it certifies retention of 72 of the 95 observed efficient portfolios. Third, we evaluate the certificate on two independently defined engineering benchmarks and connect selected manuscript interpretations to executable assertions and worked corrections.

Security portfolio simulation-optimization and necessary/possible efficiency are established ideas [2], [3]. Our contribution is their concrete use to distinguish outcome sensitivity, frontier retention, and remaining decision uncertainty in an evidence-constrained hospital experiment. The added hospital analyses are retrospective. The benchmark evaluation plan was committed before generating its candidate outcomes; the evaluations share our research workflow and do not constitute independent clinical validation.

2. Model and Evidence Design

2.1. Evidence constrains mechanisms, not every coefficient

The parameter and endpoint registries separate evidence-constrained quantities, external comparison targets, declared assumptions, model-internal quantities, and normalized scenario units. A source must match both the quantity and the population to identify a coefficient. Incident-level recovery durations, for example, do not identify per-node restoration throughput. Sector adoption figures can inform a scenario's

context without estimating the effectiveness of a deployed control.

This rule changes model structure. Patching mainly reduces exploit-mediated traversal, has little effect on credential-mediated traversal, and has partial effect on vendor pathways. Identity protection has explicit coverage and effectiveness on eligible authentication pathways. Evidence about commercial account authentication [4] is not applied as a reduction in whole-network ransomware probability. Isolated backups have no normal traversal path, but can suffer a per-incident isolation lapse or residual recovery failure. A small repeated traversal probability would instead model a delay: over T independent opportunities, a per-step probability q produces cumulative probability $1 - (1 - q)^T$.

Technical effect sizes remain non-identifiable from the available public sources. We sample 10 coefficients over declared ranges once per scenario and share that vector across candidate replays. The objectives therefore average over those ranges as well as event randomness. Guidance motivates mechanisms; it does not supply invented numerical effectiveness estimates. The complete register and mechanism specification are included in the artifact.

2.2. Network, objectives, and paired experiment

One synthetic facility is a network of digital assets with four clinical-service dependencies. Incidents enter through five categories: workstation, internet-facing asset, privileged system, medical device, or vendor connection. Spread, detection, defensive isolation, and restoration evolve in five-minute steps over 72 hours. Each service is available when its declared 60% supporting-node threshold is met. This binary digital-service construct does not measure patients treated or clinical harm. Weighted service-hours lost sum unavailable time over seven services: EHR, laboratory, pharmacy, imaging, scheduling, identity, and backup/recovery, with declared weights (1.0, 0.8, 0.9, 0.7, 0.4, 0.9, 0.6). Only the first four enter the clinical-outage endpoint; supporting-service losses still contribute to total weighted loss.

The resource-constrained, intermediate, and high-capacity profiles are fixed environments with 288, 96, and 16 distinct resolved candidates. Controls vary segmentation, patching, detection, isolation, backups, and identity protection; they cannot purchase a different profile. Each candidate has six minimized objectives: mean weighted service-hours lost, empirical upper-tail mean loss, sustained-outage frequency, non-recovery by 72 hours, implementation cost, and operational burden. The tail statistic averages losses at or above the empirical 90th percentile, including ties. Cost and burden are normalized scenario points, not monetary or staffing estimates.

Discovery contains 30 scenarios per profile and 12,000 candidate executions. Confirmation contains 400 fresh scenarios per profile and 160,000 executions. Disjoint scenario-ID blocks separate profiles and stages; within a profile, all candidates share event-keyed random inputs. Confirmation cycles through five entry categories, giving 80 scenarios per category. The estimand weights those categories equally.

The independent units are the scenarios, not the individual candidate executions, and the designed mixture is not hospital incident prevalence.

3. Decision Stability and Verification

3.1. Outcome definitions and population alignment

Let N_{pci} be the number of clinical services with an individual continuous outage longer than two hours in profile p , candidate c , scenario i . The event $Y_{pci}^{(k)} = \mathbf{1}\{N_{pci} \geq k\}$ requires at least k qualifying services; their outage intervals need not overlap. The primary endpoint uses $k = 4$. We report all four values of k and recompute the candidate-average frequency at each $k \in \{1, 2, 3, 4\}$ separately for each stage and profile. Averaging over candidates describes this enumerated design, not an incident prevalence.

For the same observations and weights, the following identity holds exactly:

$$\overline{Y^{(1)}} - \overline{Y^{(4)}} = \overline{\mathbf{1}\{1 \leq N \leq 3\}}. \quad (1)$$

The fraction of partial-service outcomes is therefore the endpoint's $k = 1$ versus $k = 4$ sensitivity, not independent evidence for its insensitivity. The identity does not justify moving a pooled discovery statistic into a profile-specific confirmation claim. We check the identity numerically and report both candidate-weighted pooling and equal-profile averaging.

3.2. Pairing, allocation, and uncertainty

For two candidates, define the paired loss difference D_{hi} in entry stratum h , with allocation weight $w_h = n_h/n$. The estimated mean-difference variance is

$$\widehat{\text{Var}}(\widehat{\Delta}) = \sum_{h=1}^5 w_h^2 \frac{s_{D,h}^2}{n_h}, \quad s_{D,h}^2 = s_{a,h}^2 + s_{b,h}^2 - 2s_{ab,h}. \quad (2)$$

Dropping covariance changes the question's uncertainty. It can either inflate or deflate an estimate depending on covariance; pairing is not guaranteed to help for every possible model. For the nonlinear tail statistic, covariance is estimated from shared stratified bootstrap draws.

We compare paired and covariance-omitting versions of the earlier "within one standard error on all four stochastic objectives" diagnostic, using the same design and bootstrap draws. This is a descriptive heuristic, not a hypothesis test, equivalence test, or proof that portfolios can be ordered. For a more explicit mean-loss comparison, every nonfree candidate is compared with its own profile's free baseline. Approximate paired t intervals use Satterthwaite degrees of freedom and Bonferroni adjustment across all 397 contrasts. This includes the observed best candidate in the family rather than presenting it as a prespecified isolated comparison. Zero observed within-stratum variance is marked unestimable, not certain.

3.3. Decision-set and specification diagnostics

A candidate is Pareto efficient if no other candidate is no worse on every included objective and strictly better on at least one. We enumerate the full finite space; no optimization heuristic approximates this set. Let $F(X)$ denote the frontier in candidate set X , and let S be the 65 candidates labeled finalists by the frozen discovery rule. We distinguish *omissions*, $F(X) \setminus S$, from *restricted-view artifacts*, $F(S) \setminus F(X)$. A shortlist can omit efficient candidates without containing any spurious efficient candidate itself.

We resample whole scenarios within each entry category, preserving all candidate and endpoint pairings. Each of 1000 draws recomputes the six objectives and both frontiers. Independent profile blocks are resampled separately. Finalist labels remain fixed: the analysis conditions on the observed discovery decision and does not rerun its selection rule. Percentile summaries are conditional resampling ranges, not confidence sets for a uniquely true frontier or forecasts for other hospitals. Discrete dominance and selection make ordinary bootstrap coverage especially uncertain.

We also enumerate all 63 nonempty subsets of the six objectives and compare frontier membership across k . Jaccard similarity is intersection size divided by union size. Finally, 500 stress draws independently multiply each of eight cost and eight burden coefficients by a $\text{Uniform}(0.5, 1.5)$ factor, shared across profiles. Draws reversing the cumulative order of the segmentation or backup tariffs are rejected (186 rejections). These are analyst-declared perturbations, not empirical priors. Positive uniform scaling of cost and burden is checked separately: it preserves Pareto dominance algebraically, although keeping a numeric budget fixed can change affordability.

3.4. An interval certificate for frontier retention

Let $f_c \in \mathbb{R}^m$ be candidate c 's minimized objective vector, with componentwise bounds $L_c \leq f_c \leq U_c$. Write $a \prec b$ when a is no worse in every coordinate and strictly better in at least one. The finite Cartesian box $\mathcal{B}$ permits independent choices across candidates and objectives. Define

$$G = \{c : \nexists d \neq c, L_d \prec U_c\}, \quad (3)$$

$$P = \{c : \nexists d \neq c, U_d \prec L_c\}. \quad (4)$$

Proposition. $G = \bigcap_{f \in \mathcal{B}} F(f)$ and $P = \bigcup_{f \in \mathcal{B}} F(f)$. Consequently, $G \subseteq F(f) \subseteq P$ for every admissible matrix.

Proof. If $L_d \prec U_c$, choosing d at its lower bound and c at its upper bound makes c dominated. Conversely, any realized dominator implies $L_d \prec U_c$, so its absence guarantees efficiency. If $U_d \prec L_c$, d dominates c in every realization. If no such d exists, put c at L_c and every competitor at its upper bound: none dominates c , proving possible efficiency. Independence permits these simultaneous corner assignments. Strictness retains exact ties. $\square$

The algorithm requires $O(n^2m)$ comparisons and returns a competing candidate for each failed guarantee or excluded possibility. These are finite-box instances of established necessary/possible efficiency concepts [3]. We use

them as an interpretable diagnostic, without a probabilistic coverage claim. If the true specification set is a correlated subset of $\mathcal{B}$, G remains sufficient for retention and P remains an outer bound; they need not be exact for that smaller set.

For the hospital application, five objectives remain fixed at their observed confirmation estimates. The outage coordinate of each candidate ranges independently from its $k = 4$ frequency to its $k = 1$ frequency. This contains all four common- k specifications and additional combinations that a common threshold cannot produce. Certifying a candidate across this entire box is stronger than observing it at four thresholds. An uncertified candidate need not actually change membership under those four thresholds.

3.5. Parameter, structure, and external transfer checks

The preserved parameter analysis compares candidate rankings between the low and high thirds of each sampled coefficient. Separately, the structural experiment contrasts restoration gated on containment with restoration proceeding in parallel. It uses a distinct paired bank of 150 scenarios per profile and 96 selected candidates (frontier members, comparators, and preference selections), rather than reenumerating the entire space. Membership comparisons are relative to that selected set and bank.

For transfer beyond the hospital objectives, we use the published RE21 four-bar truss and RE22 reinforced-concrete beam implementations [5]. We archive the unchanged upstream code, commit, hashes, and license. A committed evaluation plan specifies 128 candidates sampled uniformly over each problem's variable bounds, and independent objective intervals of $\pm 1\%$, $\pm 5\%$, and $\pm 10\%$ of each sampled objective range, with nonnegative lower bounds. RE22 retains its published constraint violation objective; not every sampled design is feasible.

Each of the six settings has 1,000 interior perturbation matrices. A separate two-objective sorting oracle checks frontier inclusion between G and P . For the first eight candidates per problem, a second check enumerates all 256 corners with only the second objective uncertain at 5% bounds, requiring exact agreement with the frontier intersection and union. These checks evaluate implementation and transfer to fixed finite benchmark sets. The stress bounds are not measured engineering uncertainty.

3.6. Executable interpretation assertions

The registry gives each selected claim a source table, scope filter, expected groups and row count, statistic, unit, reduction, quantifier, relation, and bound. For example, interpreting a claim that all profile gaps are roughly three percentage points as $\max_p \Delta_p \leq 4$ percentage points requires the confirmation population and every profile. Four is a generous audit tolerance for that wording, not a clinical cutoff. Failed assertions return group-specific witness values.

TABLE 1. ILLUSTRATIVE BUDGET-CONSTRAINED COMPARISONS. FREE AND SELECTED COLUMNS ARE MEAN WEIGHTED SERVICE-HOURS LOST. SELECTION MINIMIZES OBSERVED MEAN LOSS WITHIN THE DECLARED POINT BUDGET; THE FINAL COLUMN IS THAT CANDIDATE’S $k = 4$ EVENT FREQUENCY.

Profile	Budget	Free	Selected	Event %
Resource-constrained	5	322.13	156.99	94.50
Intermediate	10	179.34	24.34	21.50
High-capacity	15	15.02	13.07	0.00

TABLE 2. CONFIRMATION CANDIDATE-AVERAGE EVENT FREQUENCIES (%), THEIR DIFFERENCE IN PERCENTAGE POINTS, AND FRONTIER JACCARD SIMILARITY BETWEEN $k = 1$ AND $k = 4$.

Profile	$k = 1$	$k = 4$	Gap	Jaccard
Resource-constrained	87.32	83.45	3.87	0.896
Intermediate	57.39	44.79	12.60	1.000
High-capacity	16.00	1.66	14.34	1.000

Missing populations, unit mismatches, and unsupported operations are invalid and stop verification.

The finite evaluation has 8 correct assertions, 3 historical contradictions, and 8 single-field mutations of correct assertions. It includes exact-boundary, missing-population, and unsupported-operation cases. These were constructed after reviewing the paper. Detection on this set demonstrates the specified checks, not sensitivity or specificity on unseen manuscripts. A human must still judge whether an assertion captures the intended sentence and whether the source itself has scientific validity.

4. Results

4.1. Defense tradeoffs within the synthetic model

Table 1 connects the six-objective experiment to a concrete decision question. It compares each profile’s free baseline with its lowest-mean-loss candidate inside the profile’s declared budget. This one-objective selection is illustrative: another objective or burden constraint can favor another portfolio. Comparing profiles does not estimate the benefit of upgrading a hospital, because their starting environments differ and capacity is fixed. A zero observed event frequency is not a zero-risk guarantee. The artifact records the selected portfolio identifiers and their costs and burdens alongside the generated table.

4.2. Endpoint sensitivity and certified retention

The intermediate and high-capacity profiles have double-digit percentage-point changes in endpoint frequency, yet identical frontier membership across all four thresholds. The resource-constrained frontier changes despite a smaller frequency gap (Table 2). Thus frequency sensitivity does not order decision-set sensitivity. The historical claim of

TABLE 3. INTERVAL CERTIFICATE WITH FIVE OBJECTIVES FIXED AND EACH CANDIDATE’S OUTAGE FREQUENCY RANGING INDEPENDENTLY FROM $k = 4$ TO $k = 1$. G : GUARANTEED EFFICIENT; F_4 : OBSERVED FRONTIER; P : POSSIBLY EFFICIENT.

Profile	Candidates	$ G $	$ F_4 $	$ P $
Resource-constrained	288	42	64	79
Intermediate	96	21	22	41
High-capacity	16	9	9	16

uniformly small threshold effects is contradicted by these profile-specific results.

The certificate retains 72 of 95 observed frontier members (75.8%) throughout the entire declared endpoint box (Table 3). All 9 high-capacity members are guaranteed to remain efficient, but 7 other candidates could join them. This certifies retention, not exact invariance of the frontier. A possible dominator in the released candidate table explains why each remaining candidate lacks a guarantee; a necessary dominator explains exclusion from the possible set.

Pooling can conceal the profile dependence. Resource-constrained candidates contribute 72% of executions, intermediate candidates 24%, and high-capacity candidates 4%. The discovery fraction with one to three qualifying services is 3.10% pooled but 6.46% with equal profile weights; intermediate and high-capacity values are 9.03% and 9.58%. The corresponding confirmation aggregates are 6.38% and 10.27%. Equation 1 applies within each consistent population and weighting, not across these different aggregates.

4.3. Structural assumptions can change the decision

The preserved parameter analysis reports a worst low-versus-high-third rank correlation of 0.606 across its evaluated coefficients and objectives. This is descriptive evidence of varying ranking stability, not uniform robustness across the parameter space. The separate structural experiment finds 21 candidates efficient only with containment-gated restoration and 3 only with parallel restoration, within its 96 selected candidates. Intermediate mean loss rises from 46.93 to 107.41 weighted service-hours when restoration can proceed before containment.

The modeled mechanism explains the direction: restored nodes re-enter the network susceptible while an adversary may retain reach. Earlier restoration can therefore enable reinfection. The high-capacity selected frontier retains its membership under the alternative, although its objective values change slightly. Neither parameter rankings nor the endpoint certificate covers this structural change, because it changes more than the bounded outage coordinate. This distinguishes three separate claims about stability.

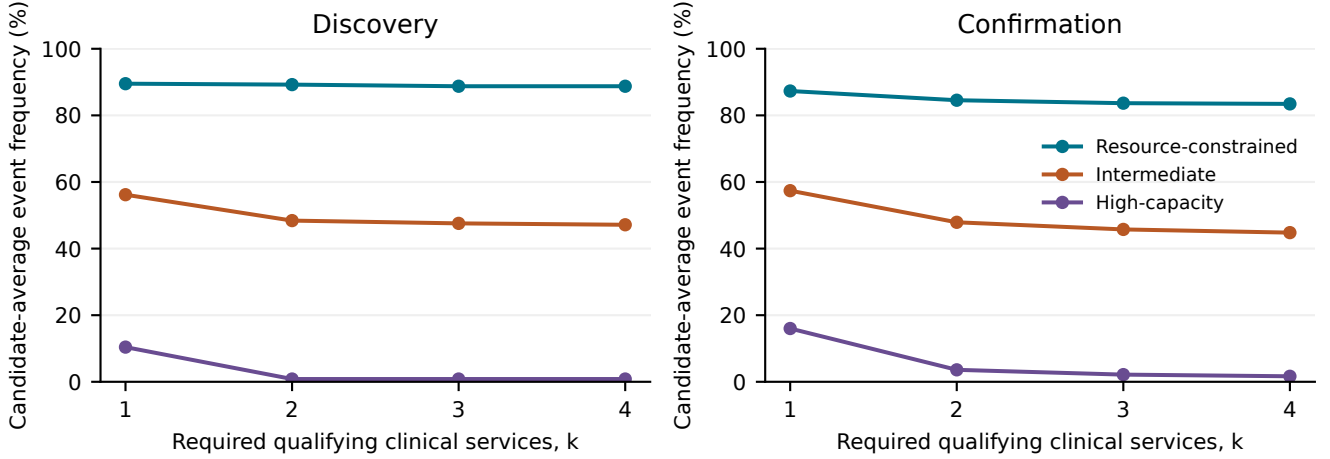


Figure 1. The nested endpoint ladder, separated by experimental stage. Each point averages the resolved candidates in one profile. These are digital-service event frequencies in the designed experiment, not hospital incident prevalence.

TABLE 4. ALL PLANNED ENGINEERING BENCHMARK SETTINGS, WITH 128 SAMPLED CANDIDATES PER PROBLEM. RADIUS IS A FRACTION OF EACH SAMPLED OBJECTIVE RANGE. F_0 IS THE NOMINAL FINITE FRONTIER.

Problem	Radius	$ G $	$ F_0 $	$ P $
RE21	1%	9	15	19
RE21	5%	0	15	65
RE21	10%	0	15	120
RE22	1%	0	12	91
RE22	5%	0	12	120
RE22	10%	0	12	125

4.4. Transfer to independently defined benchmark problems

Every one of the 6,000 perturbation frontiers contains G and lies within P . The two exact corner checks also match both the intersection and union over 512 matrices. These numerical checks support the implementation of the proposition; the proposition, rather than random sampling, establishes the full-box inclusion.

The informative negative result is visible in Table 4. RE21 retains nine guaranteed candidates at 1% bounds, while 5 of the six settings have empty guaranteed sets. Wider intervals admit many possibly efficient designs. Empty G means that no particular design is certified across every matrix; it does not mean every realized frontier is empty or that the algorithm failed. These benchmark cases demonstrate portability and the cost of broad bounds, without providing an independent hospital experiment.

4.5. Pairing changes uncertainty interpretation

Table 5 isolates the effect of covariance under the fixed entry-category allocation. Near-pair counts decrease substantially when pairing is retained. Falling outside this heuristic

TABLE 5. PAIRS MEETING THE ONE-SE PROXIMITY HEURISTIC ON ALL STOCHASTIC OBJECTIVES, WITH MATCHED STRATA AND BOOTSTRAP DRAWS. OMITTED DROPS COVARIANCE; PAIRED RETAINS IT. NEITHER COLUMN COUNTS STATISTICAL EQUIVALENCE.

Profile	All pairs	Omitted	Paired
Resource-constrained	41,328	770	88
Intermediate	4,560	84	10
High-capacity	120	32	2

TABLE 6. OBSERVED SIX-OBJECTIVE FRONTIERS. RESTRICTED USES FROZEN FINALISTS; OMISSIONS ARE FULL-SPACE EFFICIENT CANDIDATES OUTSIDE THAT SHORTLIST; ARTIFACTS ARE EFFICIENT ONLY WITHIN THE SHORTLIST.

Profile	Full	Restricted	Omitted	Artifacts
Resource-constrained	64	36	28	0
Intermediate	22	17	5	0
High-capacity	9	4	5	0

does not establish statistical ordering, but its dependence on the variance calculation makes *indistinguishability* an inappropriate label.

The high-capacity candidate with lowest observed mean loss saves 1.95 weighted service-hours against its free baseline. Its stratified paired SE is 0.333, versus 0.944 with covariance omitted. The approximate simultaneous interval over all 397 baseline contrasts is [0.66, 3.25] hours. The positive interval supports a within-model mean difference. It also contradicts the historical caption’s claimed under-one-hour vertical range: the baseline-to-best difference alone exceeds that amount.

4.6. Shortlist completeness is conditional

The full-space frontier contains 95 candidates and the restricted frontier 57, with 38 omitted efficient candidates

TABLE 7. FRONTIER SENSITIVITY TO 500 COMPONENT-WEIGHT STRESS DRAWS. SIMILARITY IS RELATIVE TO THE ORIGINAL FRONTIER; ADDED AND REMOVED COLUMNS GIVE MAXIMA ACROSS DRAWS.

Profile	Median J	Min J	Added	Removed
Resource-constrained	0.728	0.500	58	14
Intermediate	0.724	0.438	26	8
High-capacity	0.818	0.500	6	3

(Table 6). The rounded central 95% stratified-resampling ranges are 83–106 for the full frontier and 29–49 for omissions. Every resample contains an omission. The existence of omissions is stable in this conditional exercise; their exact number is not.

Although the observed bank has zero restricted-view artifacts, at least one appears in 42.7% of resamples (Figure 2). This is conditional on the frozen discovery shortlist, not the failure probability of an independently selected shortlist.

4.7. Objective and tariff choices

Heterogeneous tariffs alter membership in every profile (Table 7); uniform positive unit conversion preserves dominance. The objective enumeration also reveals why a large frontier alone is not a novelty claim. Across two-objective subsets, sizes range from 1–16, 1–18, and 1–9 in the three profiles, versus 64, 22, and 9 with all six objectives. Tied single-objective minima can be numerous: one intermediate objective has 47. Appendix B reports all dimensions.

4.8. From a sentence to a checkable conclusion

The preserved numeric audit passes on the archived paper despite all three contradictions in Table 8. Explicit assertions reject all 3 historical cases with witnesses, accept 8 correct controls, and reject or flag invalid all 8 single-field mutations as specified. The mutations cover population, unit, aggregation, bound, boundary strictness, missing data, and unsupported operations. This finite regression evaluation supplements the scientific argument with inspectable checks; it does not measure accuracy on previously unseen prose.

5. Relation to Prior Work

Kiesling et al. already combine security-control models, simulation, multi-objective optimization, and portfolio selection [2]. Our hospital-specific contribution is the explicit evidence-to-mechanism framework, clinical-service dependency outcomes, exhaustive paired finite comparison, and separation of endpoint retention from other uncertainties. Hospital process simulation with local observations and expert input [6] addresses a different validation level; our synthetic network does not inherit that empirical grounding.

Necessary and possible efficiency under interval uncertainty are established optimization concepts [3]. Equations 3–4 give a direct finite-box specialization, proof, and witness-producing implementation. The added evidence is

its endpoint application and transparent evaluation on the published RE suite [5]. The benchmark perturbations assess this implementation, not the continuous engineering problems’ global Pareto fronts.

STRESS reporting guidance [7], common random numbers [8], and the distinction between prior plans and retrospective analysis [9] inform the experimental design. Multiverse and specification-curve work [10], [11] motivate examining analytical choices; our enumeration is descriptive and does not implement specification-curve joint inference.

Statistical consistency checks [12], metamorphic simulation tests [13], and the recent ABE-Ralph experimental-fidelity preprint [14] already address relations beyond successful execution. Our small deterministic registry contributes case-specific scope, unit, and quantifier checks. Its role is to make interpretations reviewable beside the model and decision diagnostics, rather than to claim general automated scientific judgment.

6. Limitations, Ethics, and Transfer

The hospital findings concern one synthetic facility and declared parameter ranges. Binary service availability, a 72-hour horizon, and assumed control effects limit transport to clinical capacity, patient harm, regional spread, or months of recovery. The external-validation registry contains nine benchmarks: one numeric failure, one loose numeric consistency result, two qualitative failures, three not scored, and two structurally unaddressable. These comparisons do not validate procurement recommendations.

The interval certificate is exact for the declared Cartesian box, conditional on five fixed estimated objectives in the hospital application. It does not account for their Monte Carlo error or for structural uncertainty. Correlated specification sets can make its bounds conservative. The structural analysis uses selected candidates, and only one structural choice is varied. The bootstrap conditions on the observed bank and frozen shortlist; ordinary percentile coverage for discontinuous Pareto selection is not established. The family-adjusted mean intervals are approximate.

The engineering problems have independent published definitions but were evaluated in the same workflow. Their finite sampled candidates and synthetic bounds test transfer and implementation, not clinical realism or independently conducted replication. Historical assertion cases were selected after review, so they cannot establish unseen-error detection accuracy. An omitted or weak assertion can leave an incorrect sentence unchecked.

The study uses synthetic trials and public aggregate sources, with no patient records, participant recruitment, or live-system experiments. The artifact preserves failed comparisons and historical source-state limitations so that readers can distinguish a verified computational relationship from an empirically identified mechanism.

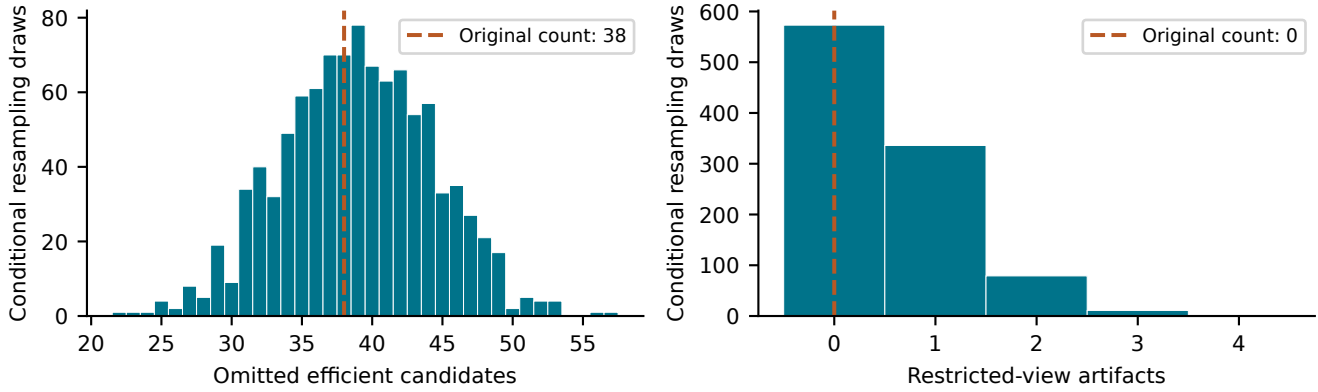


Figure 2. 1000 stratified scenario resamples, summed across independent profile banks, with finalist labels fixed. Dashed lines mark original-bank counts. These are conditional distributions, not confidence intervals for a population’s true frontier.

TABLE 8. WORKED INTERPRETATION AUDIT. HISTORICAL SENTENCES ARE SHORTENED FAITHFULLY FROM THE ARCHIVED MANUSCRIPT. ASSERTIONS ENCODE THE STATED SCOPE, UNIT, AND BOUND; A PASSING NUMERIC-MACRO CHECK ALONE DID NOT TEST THESE RELATIONS.

Historical statement	Numerical or scope evidence	Executable assertion	Corrected conclusion
Endpoint values move by roughly three percentage points across k .	High-capacity confirmation gap: 14.34 percentage points.	All confirmation profiles present; $\max_p \Delta_p \leq 4$ pp. Fails.	Frequency changes are profile-specific; unchanged frontiers do not imply small frequency gaps.
The high-capacity panel spans under one service-hour.	Baseline-to-best mean difference: 1.95 weighted service-hours.	High-capacity mean-loss range < 1 hour. Fails.	The observed baseline-to-best span exceeds one hour; paired uncertainty is reported separately.
The figure presents discovery results.	Plotted values come from the confirmation objective table.	Source stage equals discovery. Fails.	Identify confirmation as the plotted stage and report discovery separately.

7. Conclusion

The project links a pathway-specific hospital defense model to explicit evidence classes and decision-stability checks. Large changes in an endpoint frequency can co-exist with an unchanged observed frontier. The interval certificate strengthens that observation: 72 of 95 efficient portfolios are retained across the full stated endpoint box, while possible additions remain visible. Structural alternatives, paired uncertainty, and shortlist resampling expose different limitations that endpoint retention alone cannot resolve. Published engineering checks demonstrate that the certificate transfers to other finite objective tables, including cases where broad bounds are inconclusive. This yields a concrete reporting principle: state which outcome, candidate set, and uncertainty region a stability claim covers, and provide a witness when the claim fails.

Artifact Availability

The artifact includes model code, preserved trial banks and protocols, retrospective analysis plans, the benchmark plan and licensed upstream functions, all result tables, manuscript sources, and verification scripts. Compressed

raw confirmation data are restored losslessly by the supplied script. All reported results are generated from recorded data. Author identities and the identified repository link are omitted from this manuscript copy.

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LLM Usage Statement

AI tools assisted with literature search, manuscript editing, and development and checking of additional analyses.

Appendix A. Reproduction and Audit Boundary

The original raw banks are unchanged. New analysis starts from a committed, clean source state captured before output generation. Each manifest records the source commit, environment, input and output sizes, and SHA-256 hashes. The historical generated-number manifest records a dirty working tree; we preserve that fact. Hash agreement verifies artifact consistency but cannot retroactively prove that the historical code was committed before execution. A commit-attribution map resolves historical source identifiers after the repository’s metadata-only attribution correction.

For the retrospective analysis and benchmark checks, Python 3.12 and exact package versions are recorded in the manifest and revision environment file. The reproduction guide gives commands to restore the compressed bank, run the verification harness, regenerate the analyses and benchmark checks from clean checkouts, and generate numeric TeX and vector figures after committing the result tables. The paper generator’s check mode recomputes the TeX and verifies its manifest without overwriting files. These checks certify specified artifact relations, not the correctness of every sentence.

The retrospective plan records an amendment: early diagnostic output treated entry categories as randomly mixed, but source inspection established a fixed allocation. Final results preserve that allocation. Initial outputs are retained in Git history rather than relabeled as the final design. For Eq. 2, set $v_h = w_h^2 s_{D,h}^2 / n_h$ and use approximate degrees of freedom $(\sum_h v_h)^2 / \sum_h [v_h^2 / (n_h - 1)]$. The interval is $\hat{\Delta} \pm t_{1-.05/(2.397), \nu} \sqrt{\sum_h v_h}$.

Appendix B. Objective-Subset Enumeration

TABLE 9. FRONTIER SIZE OVER ALL OBJECTIVE SUBSETS OF EACH DIMENSION. ENTRIES ARE MINIMUM/MEDIAN/MAXIMUM, INCLUDING TIED OPTIMA. R, I, AND H DENOTE RESOURCE-CONSTRAINED, INTERMEDIATE, AND HIGH-CAPACITY PROFILES.

Dimensions	R	I	H
1	1/1/3	1/1/47	1/1.5/8
2	1/6/16	1/3/18	1/2/9
3	1/18.5/45	1/14/22	1/4.5/9
4	6/45/53	1/21/22	1/6/9
5	28/54/64	14/22/22	5/9/9
6	64/64/64	22/22/22	9/9/9

The single-objective intermediate maximum is caused by tied optima. Adding an objective can break such ties and remove a previously efficient candidate. Therefore frontier cardinality need not increase monotonically with dimension in this discrete problem. The enumeration characterizes this candidate set; it is not a distribution-free prediction of frontier size.